CHAPTER 9

Orthogonality and distances

S48, S51, S54, S56, S60

> **Key point · Objectives.** Calculate a dot product; justify orthogonality; find an orthogonal projection and a distance; calculate a volume.

Course notes

The coordinate calculations in this chapter are carried out in an orthonormal frame of space.

Dot product in space

For non-zero vectors $\vec{u}, \vec{v}$, $\vec{u} \cdot \vec{v} = \|\vec{u}\|\|\vec{v}\| \cos\theta$, where $\theta \in [0, \pi]$ is their angle. If one of them is zero, the dot product is 0.

$$\vec{u} \cdot \vec{v} = \tfrac{1}{2}(\|\vec{u} + \vec{v}\|^2 - \|\vec{u}\|^2 - \|\vec{v}\|^2).$$

In an orthonormal frame, for $\vec{u} = (x, y, z)$ and $\vec{v} = (x', y', z')$,

$$\vec{u} \cdot \vec{v} = xx' + yy' + zz'.$$

The dot product is symmetric and bilinear; $\vec{u} \cdot \vec{u} = \|\vec{u}\|^2$. Two vectors are orthogonal if and only if their dot product is zero.

Example. $\vec{u} = (1, 2, 2)$ and $\vec{v} = (2, -1, 0)$ are orthogonal: $2 - 2 + 0 = 0$. Their norms are 3 and $\sqrt{5}$.

Line, plane and normal vector

Two lines are orthogonal if their direction vectors are. A line is orthogonal to a plane if it is orthogonal to two non-collinear directions of that plane. A normal vector to a plane is a non-zero vector orthogonal to all its directions.

Why two directions suffice. If $\vec{n} \cdot \vec{u} = \vec{n} \cdot \vec{v} = 0$, then, for every $\vec{w} = s\vec{u} + t\vec{v}$, $\vec{n} \cdot \vec{w} = s\vec{n} \cdot \vec{u} + t\vec{n} \cdot \vec{v} = 0$.

> **Key point · Vector condition for belonging to a plane.** If A belongs to the plane $\mathcal{P}$ with normal vector $\vec{n}$, then $M \in \mathcal{P}$ if and only if $\overrightarrow{AM} \cdot \vec{n} = 0$.

Orthogonal projection and distance

The orthogonal projection H of M onto a plane is the unique point of the plane such that $\overrightarrow{MH}$ is normal to the plane; if M belongs to the plane, $H = M$. The distance from M to the plane is MH. On a line with direction vector $\vec{u}$, the projection satisfies $H \in d$ and $\overrightarrow{MH} \cdot \vec{u} = 0$.

Proof of the minimum. For every point P of the plane, $\overrightarrow{MH} \perp \overrightarrow{HP}$, so $MP^2 = MH^2 + HP^2 \geq MH^2$. Equality holds only for $P = H$. The projection is the closest point of the plane.

Method on a line. If $d = A + \mathbb{R}\vec{u}$, write $H = A + t\vec{u}$ and solve $(A - M + t\vec{u}) \cdot \vec{u} = 0$. Hence $t = \frac{\overrightarrow{AM} \cdot \vec{u}}{\|\vec{u}\|^2}$.

Angles, areas and volumes

For an angle $\widehat{ABC}$, use $\overrightarrow{BA}$ and $\overrightarrow{BC}$: $\cos \widehat{ABC} = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{BA \times BC}$. For the acute angle between two lines, take the absolute value of the quotient. The area of a triangle is $\frac{bh}{2}$; the volume of a tetrahedron is $\frac{\mathcal{A}_{\text{base}} h}{3}$, where h is an orthogonal distance. The units are squared and cubed respectively.

Expansions and loci

By bilinearity,

$$\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 + 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2, \qquad \|\vec{u} - \vec{v}\|^2 = \|\vec{u}\|^2 - 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2.$$

We can therefore also calculate $\vec{u} \cdot \vec{v} = \frac{1}{2}(\|\vec{u}\|^2 + \|\vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$. For two distinct points A, B with midpoint I, the locus of points M such that $MA = MB$ is the plane through I with normal vector $\overrightarrow{AB}$: the *perpendicular bisector plane*. Indeed, writing the vectors from I, the expansions give $MA^2 - MB^2 = 2\overrightarrow{IM} \cdot \overrightarrow{AB}$. Equality of the distances is therefore equivalent to $\overrightarrow{IM} \cdot \overrightarrow{AB} = 0$.
Extension, not assessed in 2026–2027. Two planes are perpendicular when their normal vectors are orthogonal. This property must not be confused with parallelism of planes, which corresponds to collinear normal vectors.

Research activity

The point of a line closest to a given point — 35 min.

In an orthonormal frame of space, consider $A(1,0,0)$, $M(0,2,3)$ and the line d through A with direction vector $\vec{u} = (1,1,0)$. We look for the point of d closest to M.

1. Express, in terms of a real parameter t, the coordinates of a point H of d.
2. Find the value of t for which $\overrightarrow{MH} \cdot \vec{u} = 0$. Deduce the coordinates of the resulting point H.
3. Calculate the distance MH.
4. Let P be any point of d. Prove that $MP^2 = MH^2 + HP^2$, then deduce that H is the unique point of d closest to M.

Progressive exercises

E1 · Practice. In an orthonormal basis of space, consider $\vec{u} = (2, -1, 3)$ and $\vec{v} = (1, 4, 0)$. Calculate $\vec{u} \cdot \vec{v}$.

E2 · Practice. In an orthonormal basis, consider $\vec{u} = (a, 2, 1)$ and $\vec{v} = (3, -1, 4)$, where a is real. Find a so that $\vec{u}$ and $\vec{v}$ are orthogonal.

E3 · Practice. In an orthonormal frame, a plane is spanned by $\vec{u} = (1, -1, 0)$ and $\vec{v} = (1, 0, -1)$. Show that $\vec{n} = (1, 1, 1)$ is normal to it. Then consider $A(0, 0, 0)$ and $B(2, 2, 2)$. Describe geometrically the set of points equidistant from A and B.

E4 · Practice. In an orthonormal basis, consider $\vec{u} = (1, 0, 1)$ and $\vec{v} = (0, 1, 1)$. Find the measure, between 0 and π, of the angle between these two vectors.

E5 · Practice. A tetrahedron has as its base a right-angled triangle whose two perpendicular sides measure 3 cm and 4 cm. The height of the tetrahedron relative to this base is 5 cm. Calculate its volume in cm^3.

Textbook or past-paper work

Barbazo, p. 362, exercises 23 and 27. In S48: exercise 23, to calculate a dot product in coordinates. At home after S54: exercise 27; check orthogonality to the two direction vectors of a plane, without settling for a single zero product.

Homework

HW11 — A face and its height — 50 min.

In an orthonormal frame with origin O, consider the points $A(2,0,0)$, $B(0,2,0)$ and $C(0,0,2)$.

1. Prove that the triangle ABC is equilateral and calculate its area.
2. Show that the vector $\vec{n} = (1,1,1)$ is normal to the plane (ABC).
3. Let H be the orthogonal projection of O onto (ABC). Justify that its coordinates are of the form (t,t,t). Find t using $\overrightarrow{AH} \cdot \vec{n} = 0$.
4. Calculate OH, then the volume of the tetrahedron $OABC$ taking ABC as the base. Check this volume by taking OAB as the base.

Teaching notes and solutions

> **Teacher note · Teacher notes.** Begin with the three expressions of the dot product. Orthogonal lines are not necessarily intersecting: "perpendicular" requires an intersection. To prove that a line is orthogonal to a plane, choose two non-collinear directions. The general point-to-plane distance formula is not required by heart: favour the orthogonal projection. The Cartesian equations used in the S54 progression can be introduced through the vector condition, then formalised in S67.

Solutions to the activity

1. A point H of d is written $H = A + t\vec{u}$, so $H(1+t, t, 0)$ for $t \in \mathbb{R}$.
2. $\overrightarrow{MH} = (1+t, t-2, -3)$. The dot product with $\vec{u} = (1,1,0)$ is $1+t+t-2 = 2t-1$. Setting it to zero gives $t = \frac{1}{2}$ and $H(\frac{3}{2}, \frac{1}{2}, 0)$.
3. The frame is orthonormal: $MH^2 = (\frac{3}{2})^2 + (-\frac{3}{2})^2 + (-3)^2 = \frac{27}{2}$. So $MH = \frac{3\sqrt{6}}{2}$.
4. For $P \in d$, $\overrightarrow{HP}$ is collinear with $\vec{u}$ and hence orthogonal to $\overrightarrow{MH}$. Pythagoras' theorem gives $MP^2 = MH^2 + HP^2 \geq MH^2$, with equality only if $P = H$. The point found is indeed the unique point of the line closest to M.

E1. In the orthonormal frame, the dot product is $2 \times 1 + (-1) \times 4 + 3 \times 0 = -2$.

E2. The dot product is $3a + 2(-1) + 1 \times 4 = 3a + 2$. Orthogonality is equivalent to its vanishing: $3a + 2 = 0$, so $a = -\frac{2}{3}$.

E3. The two direction vectors are non-collinear and both dot products are zero. The midpoint is $I(1,1,1)$. The required locus is the plane through I with normal vector $\overrightarrow{AB} = (2,2,2)$, hence with normal vector $(1,1,1)$.

E4. Both vectors are non-zero, with norms $\sqrt{2}$, and their dot product is 1. So their angle $\theta \in [0, \pi]$ satisfies $\cos\theta = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2}$, that is $\theta = \frac{\pi}{3}$.

E5. The area of the right-angled base is $\frac{3\times 4}{2} = 6\,\text{cm}^2$. The height perpendicular to this base is 5 cm, so $V = \frac{6\times 5}{3} = 10\,\text{cm}^3$.

Solutions to HW11

1. $\overrightarrow{AB} = (-2,2,0)$, $\overrightarrow{AC} = (-2,0,2)$ and $\overrightarrow{BC} = (0,-2,2)$; the three norms are $2\sqrt{2}$. The triangle is equilateral. Its height is $\sqrt{(2\sqrt{2})^2 - (\sqrt{2})^2} = \sqrt{6}$; its area is therefore $\frac{(2\sqrt{2})(\sqrt{6})}{2} = 2\sqrt{3}$.
2. The vectors $\overrightarrow{AB}, \overrightarrow{AC}$ are non-collinear. Their dot products with $(1,1,1)$ are respectively $-2+2 = 0$ and $-2+2 = 0$. The non-zero vector $\vec{n}$ is therefore normal to (ABC).
3. The projection of O lies on the line through O with direction vector $\vec{n}$, so $H = (t,t,t)$. Its condition for belonging to the plane is $(t-2)+t+t = 0$, that is $t = \frac{2}{3}$. We obtain $H(\frac{2}{3}, \frac{2}{3}, \frac{2}{3})$, which satisfies both conditions for an orthogonal projection.

4. $OH = \sqrt{3(\frac{2}{3})^2} = \frac{2}{\sqrt{3}}$. Taking ABC as the base, the volume is $V = \frac{(2\sqrt{3})(\frac{2}{\sqrt{3}})}{3} = \frac{4}{3}$ volume units. Other base: the triangle OAB is right-angled at O, with area $\frac{2\times 2}{2} = 2$; the line (OC) is perpendicular to the plane (OAB), and $OC = 2$. Thus $V = \frac{2\times 2}{3} = \frac{4}{3}$.

Solutions to the textbook or past-paper work

Exercise 23. In the given orthonormal frame, the dot product is the sum of the products of the corresponding coordinates:

$$(-3.5)(-4) + 2 \times 5.5 + (-4)(-3) = 14 + 11 + 12 = 37,$$

$$\frac{2}{3}\left(-\frac{3}{4}\right) + \frac{15}{7}\frac{14}{3} + \left(-\frac{12}{11}\right)\left(-\frac{22}{4}\right) = -\frac{1}{2} + 10 + 6 = \frac{31}{2}.$$

Exercise 27. In each case, the two given vectors $\vec{u}$ and $\vec{v}$ span the plane and are not collinear. A non-zero vector $\vec{w}$ is normal if, and only if, it is orthogonal to each of them.

1. $\vec{w} \cdot \vec{u} = -132 + 312 - 180 = 0$ and $\vec{w} \cdot \vec{v} = -66 + 546 - 480 = 0$. The vector $\vec{w}$ is normal to the plane.
2. $\vec{w} \cdot \vec{u} = 20 + 6 - 26 = 0$, but $\vec{w} \cdot \vec{v} = 18 - 6 - 10 = 2 \neq 0$. It is not normal to the plane.
3. $\vec{w} \cdot \vec{u} = \frac{1}{2} + 6 - 7 = -\frac{1}{2} \neq 0$. This single non-zero product is enough to rule out normality.